

Maximum Entropy Spectral Estimation

Script: Chapter 11 (AR models, Yule–Walker, Levinson–Durbin, Burg)

with solutions**Learning goals**

- Describe a resonance by an autoregressive (AR) model and relate the AR poles to resonance frequency and damping.
- Derive and solve the Yule–Walker equations with the Levinson–Durbin recursion; implement Burg’s method; explain the maximum entropy principle.
- Compare MEM (AR) spectra with windowed periodograms on the very short records available in a knock window (32–128 samples): frequency accuracy, peak shape, resolution, artefacts.
- Select the model order (FPE, AIC, MDL) and recognise the symptoms of a wrong order (spurious peaks, line splitting).

Background

A knock window is short: 60° crank angle are 3.3 ms at 3000 rpm and only 1.7 ms at 6000 rpm, i.e. 42–83 samples at $f_s = 25$ kHz (the rate of Sheet 10). The DFT of N samples has the bin spacing f_s/N (781 Hz for $N = 32$), and windowing widens the peaks further. Parametric methods fit a model to the data and are not bound to this “Rayleigh limit”.

An AR(p) process is white noise $e[n]$ (variance σ^2) filtered by an all-pole filter:

$$x[n] = -\sum_{i=1}^p a_i x[n-i] + e[n], \quad A(z) = 1 + \sum_{i=1}^p a_i z^{-i}, \quad P(f) = \frac{\sigma^2 T_s}{|A(e^{j2\pi f/f_s})|^2}.$$

Multiplying by $x[n-m]$ and taking expectations gives the *Yule–Walker equations* $\sum_{i=1}^p a_i r[m-i] = -r[m]$, $m = 1, \dots, p$, and $\sigma^2 = r[0] + \sum_{i=1}^p a_i r[i]$. The *Levinson–Durbin recursion* solves them order by order in $O(p^2)$ operations ($E_0 = r[0]$):

$$k_m = -\frac{r[m] + \sum_{i=1}^{m-1} a_i^{(m-1)} r[m-i]}{E_{m-1}}, \quad a_i^{(m)} = a_i^{(m-1)} + k_m a_{m-i}^{(m-1)}, \quad a_m^{(m)} = k_m, \quad E_m = (1 - k_m^2) E_{m-1}.$$

Burg’s method uses the same order recursion, but estimates every reflection coefficient k_m directly from the data by minimising the sum of forward and backward prediction error powers,

$$k_m = -\frac{2 \sum_{n=m}^{N-1} f_{m-1}[n] b_{m-1}[n-1]}{\sum_{n=m}^{N-1} (f_{m-1}^2[n] + b_{m-1}^2[n-1])}, \quad \begin{aligned} f_m[n] &= f_{m-1}[n] + k_m b_{m-1}[n-1], \\ b_m[n] &= b_{m-1}[n-1] + k_m f_{m-1}[n], \end{aligned}$$

with $f_0[n] = b_0[n] = x[n]$. No autocorrelation estimate (and hence no implicit windowing of the data) is needed. Order selection criteria (E_p = prediction error power of order p , N = record length):

$$\text{FPE}(p) = E_p \frac{N+p+1}{N-p-1}, \quad \text{AIC}(p) = N \ln E_p + 2p, \quad \text{MDL}(p) = N \ln E_p + p \ln N.$$

Code: `code/jslinux/sheet11/`: `ar.h` (AR library), `fft.h` (radix-2 FFT with the API of Sheet 8, `fft(re, im, n, inverse)` – you may use your own), `memspec.c` (Exercise 11.2), `resolve.c` (Exercise 11.3).

Exercise 11.1 AR models and the maximum entropy principle**PAPER**

- a) AR(1): derive a_1 and σ^2 from $r[0], r[1]$ and the PSD. For which sign of a_1 is the process low-pass, for which high-pass?
- b) AR(2): write down the Yule–Walker equations and solve them for a_1, a_2 in terms of $\rho_1 = r[1]/r[0], \rho_2 = r[2]/r[0]$. Then run the Levinson–Durbin recursion by hand for the normalised autocorrelation $r = [1, -0.0627, -0.8973]$ and give $k_1, k_2, a_1, a_2, E_1, E_2$.
- c) Show that an AR(2) model with complex poles $z_{1,2} = re^{\pm j\theta}$ has $a_1 = -2r \cos \theta, a_2 = r^2$ and describes a resonance. Compute r, θ, a_1, a_2 for the knock mode ($f_1 = 6.5$ kHz, decay time $\tau = 0.8$ ms, $f_s = 25$ kHz) and compare with b). Estimate the -3 dB bandwidth of the AR(2) peak from r .
- d) *Maximum entropy*. Given $r[0], \dots, r[p]$, there are infinitely many PSDs consistent with these values. Burg proposed to choose the one with maximum entropy rate $h = \frac{1}{4\pi} \int_{-\pi}^{\pi} \ln P(\omega) d\omega$ (+const.). Show with a Lagrange argument that this PSD has the form $P(\omega) = 1 / \sum_{|m| \leq p} \lambda_m e^{-j\omega m}$, i.e. it is an AR(p) spectrum. How does the MEM spectrum treat the unknown lags $|m| > p$, compared with a (windowed) periodogram?
- e) Show that Burg’s reflection coefficients always satisfy $|k_m| \leq 1$. Why does this guarantee a stable (minimum-phase) $A(z)$ and why is that important? Does the Yule–Walker method with the *biased* autocorrelation estimate have the same property?

Solution

a) $r[1] = -a_1 r[0] \Rightarrow a_1 = -\rho_1, \sigma^2 = r[0] + a_1 r[1] = r[0](1 - \rho_1^2), P(f) = \sigma^2 T_s / |1 + a_1 e^{-j\omega}|^2 = \sigma^2 T_s / (1 + a_1^2 + 2a_1 \cos \omega)$. For $a_1 < 0$ (pole $z = -a_1 > 0$, positive correlation) the PSD is largest at $\omega = 0$: low-pass; for $a_1 > 0$ (pole on the negative real axis) high-pass.

b) $\begin{pmatrix} r_0 & r_1 \\ r_1 & r_0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = -\begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$, hence

$$a_1 = -\frac{\rho_1(1 - \rho_2)}{1 - \rho_1^2}, \quad a_2 = -\frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}.$$

Levinson: $E_0 = 1; k_1 = -\rho_1 = 0.0627, a_1^{(1)} = 0.0627, E_1 = (1 - k_1^2) = 0.99607; k_2 = -(r_2 + a_1^{(1)} r_1)/E_1 = -(-0.8973 - 0.00393)/0.99607 = 0.9048; a_2 = k_2 = 0.9048, a_1 = a_1^{(1)} + k_2 a_1^{(1)} = 0.0627 \cdot 1.9048 = 0.1194; E_2 = (1 - k_2^2)E_1 = 0.1806$. The prediction error drops from 1 to 0.18: the process is well predictable (narrow resonance).

c) $A(z) = (1 - z_1 z^{-1})(1 - z_2 z^{-1}) = 1 - 2r \cos \theta z^{-1} + r^2 z^{-2}$. The impulse response of $1/A(z)$ is $\propto r^n \sin((n+1)\theta)$: a damped oscillation with $\theta = 2\pi f/f_s$ and $r = e^{-T_s/\tau}$. Knock mode: $\theta = 2\pi \cdot 0.26 = 93.6^\circ, r = e^{-0.05} = 0.9512, a_1 = -2 \cdot 0.9512 \cos 93.6^\circ = 0.1195, a_2 = 0.9048$ – exactly the values of b): the autocorrelation given there is that of the knock resonance excited by white noise. For r close to 1 the -3 dB bandwidth is $\Delta f \approx (1 - r)f_s/\pi = 388$ Hz, in agreement with $1/(\pi\tau) = 398$ Hz (Sheet 10).

d) Maximise $\int \ln P d\omega$ subject to $\frac{1}{2\pi} \int P(\omega) e^{j\omega m} d\omega = r[m], |m| \leq p$. With Lagrange multipliers λ_m the variation with respect to $P(\omega)$ gives $1/P(\omega) - \sum_{|m| \leq p} \lambda_m e^{j\omega m} = 0$ (up to constants), so $1/P(\omega)$ is a non-negative trigonometric polynomial of degree p . By the Fejér–Riesz theorem it can be factorised as $|A(e^{j\omega})|^2/\sigma^2$ with a minimum-phase A of degree p : the maximum entropy PSD is the AR(p) spectrum, and the constraints are exactly the Yule–Walker equations. The MEM spectrum *extrapolates* the autocorrelation beyond lag p in the “most random” way consistent with the known lags (by the AR recursion). The periodogram implicitly sets all lags $|m| \geq N$ to zero and tapers the others (triangular/window lag weighting); this truncation causes the sinc main-lobe width $\approx f_s/N$ and the side lobes (leakage). MEM has neither, hence its sharp peaks from short records.

e) With $F = \sum f^2, B = \sum b^2, C = \sum fb$: Cauchy–Schwarz gives $|C| \leq \sqrt{FB} \leq \frac{1}{2}(F + B)$ (geometric $\leq$ arithmetic mean), so $|k_m| = 2|C|/(F + B) \leq 1$. In the Levinson structure $|k_m| < 1$

for all m is equivalent to all zeros of $A(z)$ inside the unit circle (Schur–Cohn), i.e. the synthesis filter $1/A(z)$ is stable, the prediction errors $E_m = (1 - k_m^2)E_{m-1}$ stay positive and the spectrum is well defined. The biased autocorrelation sequence is positive semidefinite, so Yule–Walker with the biased estimate also yields $|k_m| \leq 1$; the unbiased estimate (division by $N - m$) does not guarantee this and may produce unstable models.

Exercise 11.2 MEM vs. periodogram on short knock windows

JSLINUX

```
1 gcc -O2 -o memspec memspec.c -lm && ./memspec [N] [p] # default N = 64, p = 8
```

- Complete `ar_levinson()`, `ar_burg()` and `ar_psd()` in `ar.h`. Test them with the AR(2) example of 11.1 b) (you must get $a_1 = 0.1194$, $a_2 = 0.9048$).
- Part (a) of `memspec` takes one knock window (N samples starting 10° after TDC, 3000 rpm, mean removed) and computes the Hann periodogram (zero padded to 1024), the Yule–Walker and the Burg spectrum. Plot `mem.csv` for $N = 32, 64, 128$ and describe the differences.
- Part (b) estimates the knock frequency (maximum between 5.5 and 7.5 kHz) in 300 windows for $N = 32, 64, 128$ with four methods and reports mean, standard deviation, rms error, the -3 dB width of the peak and how often the second mode (10.5 kHz) is found. Interpret the table: in which respect is MEM better than the periodogram, in which not? Why does Burg’s frequency estimate scatter more than the periodogram’s?
- Complete `ar_crit()`. Part (c) applies FPE, AIC and MDL to Burg models up to order 20. Which orders are chosen, and why are about 6–8 poles plausible for the knock signal?
- Part (d) uses Burg with orders 2 to 32 for $N = 64$. Explain the growing number of peaks and the “split” knock peak, and give a rule of thumb for the order.

Solution

a) Levinson–Durbin (`code/solutions/jslinux/sheet11/ar.h`):

```
1  for (m = 1; m <= p; m++) {
2      double acc = r[m];
3      for (i = 1; i < m; i++) acc += a[i] * r[m - i];
4      k[m] = -acc / E[m - 1];
5      for (i = 1; i < m; i++) tmp[i] = a[i] + k[m] * a[m - i];
6      for (i = 1; i < m; i++) a[i] = tmp[i];
7      a[m] = k[m];
8      E[m] = (1.0 - k[m] * k[m]) * E[m - 1];
9      if (fabs(k[m]) >= 1.0) return -1;
10 }
```

Burg:

```
1  for (m = 1; m <= p; m++) {
2      double num = 0.0, den = 0.0;
3      for (n = m; n < N; n++) {
4          num += f[n] * b[n - 1];
5          den += f[n] * f[n] + b[n - 1] * b[n - 1];
6      }
7      k[m] = den > 0.0 ? -2.0 * num / den : 0.0;
8      /* update errors: go backwards in n so b[n-1] is still the old value */
9      for (n = N - 1; n >= m; n--) {
10         double fo = f[n];
11         f[n] = fo + k[m] * b[n - 1];
12         b[n] = b[n - 1] + k[m] * fo;
13     }
14     /* Levinson update of the polynomial */
15     for (i = 1; i < m; i++) tmp[i] = a[i] + k[m] * a[m - i];
16     for (i = 1; i < m; i++) a[i] = tmp[i];
```

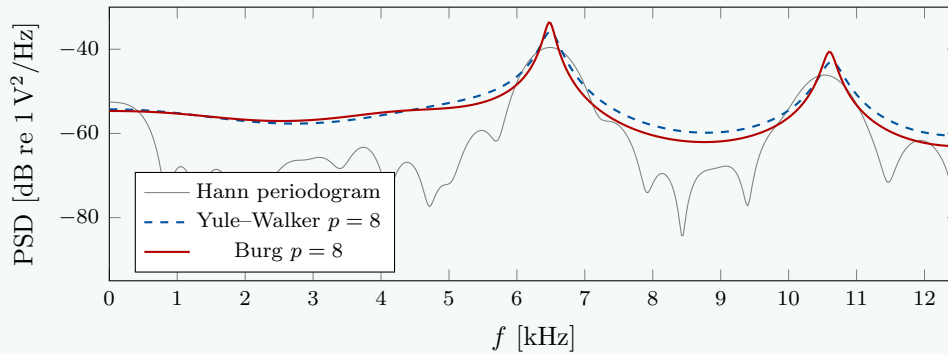
```

17     a[m] = k[m];
18     E[m] = (1.0 - k[m] * k[m]) * E[m - 1];
19 }

```

Note the backward loop over n : $b[n-1]$ must still hold the error of order $m-1$ when $f[n]$ and $b[n]$ are updated. With $r = [1, -0.0627, -0.8973]$ `ar_levinson` returns $a_1 = 0.1194$, $a_2 = 0.9048$, $E_2 = 0.181$.

b) Example window ($N = 64$, 2.56 ms, burst onset at 14.3° ATDC, $p = 8$): peak frequencies 6490 Hz (Hann periodogram), 6491 Hz (Yule–Walker), 6479 Hz (Burg); second mode at 10527/10634/10602 Hz.



The periodogram shows the two modes as broad main lobes (Hann, $N = 64$: main lobe $4f_s/N = 1.56$ kHz wide) with side lobes and a deep, ragged background. The AR spectra are smooth, have narrow peaks at both modes and a low-frequency peak (the combustion vibration that is not removed by subtracting the mean). Yule–Walker is smoother and its peaks are lower and broader than Burg’s: the biased autocorrelation estimate corresponds to an implicitly windowed (“pre- and post-windowed”) record. For $N = 32$ the periodogram lobes become twice as wide, while the Burg peaks remain sharp; for $N = 128$ all methods agree well.

c) Output of part (b):

N	method	mean [Hz]	std [Hz]	rms err [Hz]	−3 dB width [Hz]	10.5 kHz found
32	periodogram rect	6469	99	103	1043	83 %
32	periodogram Hann	6451	130	139	1540	75 %
32	Yule–Walker $p = 8$	6516	74	76	630	72 %
32	Burg $p = 8$	6487	106	107	376	74 %
64	periodogram rect	6494	21	22	519	100 %
64	periodogram Hann	6496	20	21	666	100 %
64	Yule–Walker $p = 8$	6515	37	40	382	86 %
64	Burg $p = 8$	6486	85	86	277	90 %
128	periodogram rect	6493	34	35	417	100 %
128	periodogram Hann	6496	35	35	377	98 %
128	Yule–Walker $p = 8$	6542	39	57	464	90 %
128	Burg $p = 8$	6532	84	90	354	89 %

Peak shape: the true knock spectrum has a -3 dB width of $1/(\pi\tau) \approx 400$ Hz. Burg reproduces it already from 32 samples (376 Hz); the periodograms show the width of their window main lobe (1043 Hz and 1540 Hz for $N = 32$) and approach the true width only for $N = 128$. This is the advantage of MEM: the spectral *shape* and the resolution are not limited by f_s/N .

Frequency of an isolated, strong peak: here the (zero-padded, interpolated) periodogram maximum is as good or better (rms error 21 Hz for $N = 64$); for a single sinusoid in white noise it is the maximum-likelihood estimator. A wide main lobe does not shift the maximum. Burg’s estimate scatters by 80–100 Hz: (i) its frequency estimate of a sinusoid depends on the initial phase (known bias of the Burg method for short records), (ii) the forward–backward averaging

assumes a stationary signal, but the knock burst decays – backwards it grows, so the fitted poles are pushed towards the unit circle and their angles are disturbed; (iii) with $p = 8$ the model has to share its poles between two modes, the low-frequency vibration and the noise. A noise-free test (single damped sinusoid, $N = 64$, $p = 2$) confirms (i)/(ii): Burg finds 6300–6698 Hz depending on the phase and $r \approx 0.995$ instead of 0.951. For knock *detection* (Sheet 12) the exact frequency is irrelevant; what matters is the energy near 6.5 kHz relative to the background – and there MEM’s clean, leakage-free spectrum of a 32-sample window is attractive.

d) Order selection (Burg, orders 1–20, 300 windows):

N	FPE: mean / most frequent	AIC	MDL
32	7.2 / 7	7.8 / 7	5.0 / 3
64	8.8 / 7	8.9 / 7	6.8 / 7
128	10.1 / 7	10.2 / 7	7.2 / 7

The signal contains two resonances (2 pole pairs), the low-frequency combustion vibration (1 pair) and noise: 6–8 poles are physically plausible, and all criteria mostly choose $p = 7$. FPE and AIC are asymptotically equivalent; they penalise the order weakly and tend to overestimate it for larger N (mean 10). MDL penalises with $\ln N$ and is consistent; for $N = 32$ it often selects only $p = 3$ (one resonance), because a short record does not justify more parameters.

e) Burg, $N = 64$:

p	peaks within 20 dB of max	knock peak split	rms error [Hz]
2	1.00	0 %	479
4	1.00	0 %	128
8	2.08	0 %	80
16	3.66	17 %	97
24	5.89	30 %	100
32	8.42	52 %	111

With $p = 2$ one pole pair has to describe everything (the estimate is pulled away from 6.5 kHz). For $p \gg$ (number of components) the surplus poles are placed close to the unit circle to fit noise details of the particular record: spurious peaks appear, and the knock peak splits into two (*line splitting*, typical for Burg at high order and high SNR). At $p = N/2$ the AR spectrum approaches the erratic behaviour of the periodogram (the model has as many parameters as there are independent lags). Rule of thumb: $p \approx 2 \times$ (number of expected resonances) + 2, and $p \leq N/3$; check with FPE/AIC/MDL.

Exercise 11.3 Resolving two close resonances

JSLINUX

A temperature-dependent shift of the knock mode in one cylinder, or a second structural mode of the engine block, can create two resonances only a few hundred Hz apart. Can the ECU still separate them within one knock window?

```
gcc -O2 -o resolve resolve.c -lm && ./resolve 500 && ./resolve 250 && ./resolve 1000
```

- Complete `resolved()`: the two components count as resolved if there is a local maximum within $\Delta f/2$ of f_1 and one within $\Delta f/2$ of f_2 , and the spectrum dips by at least 3 dB between them.
- The program generates two sinusoids at $f_1 = 6.5$ kHz and $f_1 + \Delta f$ (random phases) in white noise with an SNR of 40, 20 and 10 dB per component and records how often the rectangular periodogram, the Hann periodogram and Burg ($p = N/3$, at most 32) resolve them for $N = 16 \dots 256$. Determine the minimum record length for $\geq 90\%$ resolution for $\Delta f = 500$ Hz and compare with the Rayleigh limit $N \approx f_s/\Delta f$. How does the SNR influence the result?

- c) Repeat for $\Delta f = 250$ and 1000 Hz. At $\Delta f = 500$ Hz and 6000 rpm, a 60° window has 42 samples: which method can resolve the two modes?
- d) Part (b) of the program uses *damped* modes with $\tau = 0.8$ ms, as in a real knock burst. Explain why now even long records and high SNR do not help for $\Delta f = 250$ Hz and why Burg's success rate saturates at about 50–60 % for $\Delta f = 500$ Hz.

Solution

a) See code/solutions/jslinux/sheet11/resolve.c:

```
1 static int resolved(const double *P, int nf, double dfr)
2 {
3     double f2 = F1 + df_g, lo = F1 - df_g, hi = f2 + df_g;
4     int i, i1 = -1, i2 = -1, k0 = (int)(lo / dfr), k1 = (int)(hi / dfr), kmid, imin;
5     if (k1 > nf - 2) k1 = nf - 2;
6     for (i = k0 + 1; i <= k1; i++) {
7         double f = i * dfr;
8         if (P[i] > P[i - 1] && P[i] >= P[i + 1]) { /* local max */
9             if (fabs(f - F1) < 0.5 * df_g && (i1 < 0 || P[i] > P[i1])) i1 = i;
10            if (fabs(f - f2) < 0.5 * df_g && (i2 < 0 || P[i] > P[i2])) i2 = i;
11        }
12    }
13    if (i1 < 0 || i2 < 0) return 0;
14    imin = i1;
15    for (kmid = i1; kmid <= i2; kmid++) if (P[kmid] < P[imin]) imin = kmid;
16    return P[imin] < 0.5 * (P[i1] < P[i2] ? P[i1] : P[i2]); /* 3 dB dip */
17 }
```

b) $\Delta f = 500$ Hz, percentage of resolved trials (200 trials each):

N	p	SNR 40 dB			SNR 20 dB			SNR 10 dB			damped, 40 dB		
		rect	Hann	Burg	rect	Hann	Burg	rect	Hann	Burg	rect	Hann	Burg
16	5	0	0	32	0	0	1	0	0	0	0	0	28
24	8	0	0	60	0	0	22	0	0	0	0	0	34
32	10	4	0	84	2	0	58	0	0	6	1	0	44
48	16	37	10	100	39	11	100	37	4	92	31	6	46
64	21	56	25	100	49	21	100	56	24	100	34	26	50
96	32	100	50	100	100	46	100	100	46	100	39	42	58
128	32	100	100	100	100	100	100	100	100	100	36	54	58
256	32	100	100	100	100	100	100	100	100	100	35	100	58

Minimum N for $\geq 90\%$: periodogram rect 96, Hann 128 (independent of the SNR), Burg 48 at 40 dB and 20 dB (84 % already at $N = 32$ for 40 dB), 48 at 10 dB. The Rayleigh limit $f_s/\Delta f = 50$ samples is the record length at which the main lobes (width $2f_s/N$ for rect, $4f_s/N$ for Hann between the first zeros) start to separate; with random phases and the 3-dB criterion about twice this length is needed (rect: 56 % at $N = 64$, Hann: about $2\times$ longer). The periodogram's resolution is a property of the window and does not improve with SNR. Burg beats the Rayleigh limit by a factor 2 at moderate SNR – the “super-resolution” of MEM –, but its advantage shrinks with decreasing SNR: at 10 dB it needs $N = 48$ and fails completely below $N = 32$, because the noise has to be modelled by the same few poles.

c) $\Delta f = 250$ Hz (Rayleigh limit 100 samples): rect needs $N = 256$, Hann 256, Burg 64 (94 %) at 40 dB, 96 at 20 and 10 dB. $\Delta f = 1000$ Hz (Rayleigh 25): rect $N = 48$, Hann 64, Burg 24 at 40/20 dB and 32 (99 %) at 10 dB. With 42 samples (60° at 6000 rpm) and $\Delta f = 500$ Hz, only Burg can separate the modes (and only at SNR $\gtrsim 10$ dB); the periodogram resolves them in less than 40 % of the cases.

d) A damped mode has a Lorentzian line of width $1/(\pi\tau) = 400$ Hz – this is a property of the signal, not of the estimator. Two such lines 250 Hz apart overlap into a single peak, so no method can show two maxima: the success rate stays at 0–28 % for all N (output of `./resolve 250,`

damped part). At $\Delta f = 500$ Hz the lines are just separable; a long rectangular window does not help either (the signal has decayed after $\approx 3\tau = 60$ samples, more samples only add noise), the Hann window at $N = 256$ succeeds because it suppresses the discontinuity at the burst onset. Burg saturates at 50–60 %: the forward–backward error criterion assumes stationarity, so for decaying signals it biases the pole radii towards 1 and the pole angles depending on the phases (cf. 11.2 c). For damped exponentials, methods that model the decay explicitly (forward linear prediction / covariance method, Prony) are more appropriate. For $\Delta f = 1000$ Hz Burg resolves the damped modes in 90–98 % of the cases from $N = 24$ on.

Deliverables

11.1 on paper. Completed `ar.h`; plots of `mem.csv` for $N = 32, 64, 128$; the tables of `memspec` parts (b)–(d) and of `resolve` for $\Delta f = 250, 500, 1000$ Hz with a half-page interpretation: when would you use MEM in a knock controller, and when not?